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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING AY 2025-26

SERVICE LEARNING PROJECT SYNOPSIS

ON

**“Saarthi: An AI-Powered Teaching Copilot for Personalized,
Differentiated Learning Support”**

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE & ENGINEERING

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NGO Name: A RAY OF HOPE CHARITABLE TRUST

INTRODUCTION

Children supported by A Ray of Hope Charitable Trust receive vital in-person tutoring from volunteers, but those volunteers often work with limited preparation time and classrooms where students understand concepts at very different paces. A single one-hour session may include children who need extra visual support alongside others who are ready for more challenging problems, leaving little time to plan for both.

This project presents Saarthi, an AI-powered teaching copilot built specifically for NGO volunteers like this one. The platform helps a volunteer move from class analysis, to personalized lesson generation, to interactive assessment, to session analysis, and finally to a clear recommendation for the next session — all through a single workflow. Saarthi does not replace the volunteer; it gives their limited preparation time a running start.

PROBLEM STATEMENT

Despite consistent volunteer support, students at the Trust often face:

- **Lack of Differentiated Content:** Volunteers rarely have time to prepare separate materials for children at different learning levels within the same session.
- **Limited Visibility into Progress:** There is no structured way to track what each child has learned, where they are struggling, or what to revisit next.
- **Inconsistent Session Planning:** Lesson quality depends heavily on how much preparation time an individual volunteer had that week.
- **No Continuity Between Volunteers:** When different volunteers teach the same class over time, progress notes and context are often lost between sessions.

These issues limit how effectively a volunteer's limited time can be used, and make it difficult to give every child a pace of learning suited to them.

OBJECTIVE

The main goal is to develop an AI-assisted teaching copilot that helps NGO volunteers turn limited preparation time into personalized, differentiated lesson plans and a clear plan for what to teach next.

The specific objectives are:

- **Class Analysis:** Automatically analyze existing student performance to identify learning gaps and classify students by their current level of understanding.
- **Differentiated Lesson Generation:** Generate lesson plans and activities pitched at multiple difficulty levels from a single topic, so every child works on something that fits.
- **Resource-Grounded Content:** Ground AI-generated lesson content in an approved educational resource library using retrieval-augmented generation (RAG), reducing reliance on generic or unverified output.

- **Session Tracking:** Let volunteers log activities and assessment responses during a session without breaking the flow of teaching.
- **Next-Session Recommendations:** Summarize each session's outcomes and recommend what to focus on next time, carrying continuity forward even between different volunteers.

REQUIREMENTS

1. Authentication & User Roles

- Basic login system for volunteers.
- Volunteer access scoped to their assigned classes and sessions.

2. Class & Session Management

- **Class Selection:** Choose the class or section being taught.
- **Session Configuration:** Set subject, topic, duration, language, and available resources.

3. AI-Assisted Lesson Planning

- **Class Analysis:** Classify students by level using recent performance data.
- **Resource Retrieval:** Retrieve relevant approved content via FAISS-based RAG before generating a lesson.
- **Differentiated Activities:** Generate activities at multiple difficulty levels for the same underlying concept.

4. Interactive Sessions & Assessment

- Conduct activities and assessments directly through the platform.
- Record student responses — score, time taken, and attempt count — as the session happens.

5. Session Intelligence & Progress Tracking

- Summarize class performance after each session.
- Identify areas requiring reinforcement.
- Recommend what to teach next, per student and per class.

6. Content Management

- Maintain an approved educational resource library used to ground lesson generation.

7. UI/UX Design

- Web-based, responsive design usable on volunteers' own devices.
- Clear, distraction-free layouts that prioritize quick session preparation over feature depth.

TECH STACK

Layer	Technology
Frontend	React / Next.js (TypeScript)
Backend	FastAPI (Python)
Database	PostgreSQL + SQLAlchemy
AI / Retrieval	SentenceTransformers + FAISS (RAG)
Lesson Generation	LLM-based structured generation
Infrastructure	Docker, Git / GitHub

SUMMARY

This project aims to make every volunteer's limited teaching time at A Ray of Hope Charitable Trust go further by pairing them with an AI copilot for lesson planning, differentiated activities, and session intelligence. By grounding generated content in approved resources and tracking each child's progress across sessions, Saarthi helps volunteers give every child a pace of learning suited to them — even as different volunteers rotate through the same classroom.

As an individual, service-learning contribution, this project is being developed end to end — from problem identification through a working prototype — with the current implementation focused on Mathematics and Fractions, and the underlying workflow designed to extend to additional subjects as the Trust's needs grow.